SE444: Introduction to Artificial Intelligence
Mock Final Exam — Fall 2025

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Date: TBD, January 2026 Time: 120 minutes

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Show your work – any sign of serious effort will be considered.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	CLOs
I MCQ	BN Structure / d-Sep	4m	1		1
	Bayes' Rule	4m	1		1
	Decision Theory / MDPs	4m	1		1,2
	Game Theory	4m	1		1,3
	Planning (STRIPS)	4m	1		1,2
	Project Code	4m	1		4
II BN	BN + d-Separation	12m	4		1
	Bayesian Inference	12m	4		2
	MEU Decision	9m	3		2,3
III MDP	Value Iteration	15m	5		2
IV Plan/GT	STRIPS Planning	9m	3		2
	Nash Equilibrium	9m	3		3
V Code	Search Algorithms	6m	3		4
	Bayesian Updates	6m	2		4
	TOTAL	102m	**30**		

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 6 points **Total Time:** 24 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Question I.1: Bayesian Network Structure & d-Separation

Time: 4 min **Score:** 1 pt **CLOs:** CLO 1

Consider the Bayesian Network: $A \to C \leftarrow B$, $C \to D$ (where C is a collider).
Which conditional independence statements are TRUE?

A) $A \perp B$ (unconditionally)

B) $A \perp B \mid C$

C) $A \perp B \mid D$

D) $A \perp D \mid C$

E) $C \perp D$ (unconditionally)

Question I.2: Bayes' Rule Application

Time: 4 min **Score:** 1 pt **CLOs:** CLO 1

A cybersecurity system: $P(\text{Intrusion}) = 0.01$, $P(\text{Alarm}|\text{Intrusion}) = 0.95$, $P(\text{Alarm}|\neg\text{Intrusion}) = 0.05$. The alarm triggers. Which are TRUE?

A) $P(\text{Intrusion}|\text{Alarm}) > 0.5$

B) $P(\text{Intrusion}|\text{Alarm}) < P(\text{Alarm}|\text{Intrusion})$

C) The base rate significantly affects the posterior

D) If $P(\text{Intrusion})$ increases to 0.10, then $P(\text{Intrusion}|\text{Alarm}) > 0.5$

E) Bayes' rule is unnecessary; $P(\text{Intrusion}|\text{Alarm}) = P(\text{Alarm}|\text{Intrusion})$

Question I.3: Decision Theory & MDPs

Time: 4 min **Score:** 1 pt **CLOs:** CLO 1, 2

In an MDP with discount factor $\gamma = 0.9$, which statements are TRUE?

A) If $\gamma = 0$, the agent only considers immediate rewards

B) Bellman equation $V(s) = \max_a[R(s,a) + \gamma \sum_{s'} P(s'|s,a)V(s')]$ converges for $0 \leq \gamma < 1$

C) Value iteration and policy iteration produce the same optimal policy

D) Higher γ (closer to 1) makes the agent more "farsighted"

E) In deterministic MDPs, optimal policy is the same regardless of γ

Question I.4: Game Theory — Nash Equilibrium

Time: 4 min **Score:** 1 pt **CLOs:** CLO 1, 3

Consider the game (payoffs Row, Col):

	Left	**Right**
Up	(3, 3)	(1, 4)
Down	(4, 1)	(2, 2)

Which are TRUE?

A) Both players have a strictly dominant strategy

B) (Up, Left) is a Nash Equilibrium

C) (Down, Right) is a Nash Equilibrium

D) "Down" strictly dominates "Up" for Row player

E) The Nash Equilibrium yields lower payoffs for both players than (Up, Left)

Question I.5: Classical Planning — STRIPS

Time: 4 min **Score:** 1 pt **CLOs:** CLO 1, 2

Consider the following STRIPS action schema:

```
Action: Load(cargo, truck, loc)
  Precond: At(cargo, loc), At(truck, loc), ¬In(cargo, truck)
  Add:     In(cargo, truck)
  Delete:  At(cargo, loc)
```

Which statements are TRUE?

A) Forward search starts from initial state and applies actions until goal

B) Closed World Assumption means unmentioned atoms are false

C) After Load(C1,T1,Depot), At(C1,Depot) is still true

D) Backward search can be more efficient when goal is specific

E) STRIPS assumes deterministic, fully observable effects

Question I.6: Project Code — Search Algorithms

Time: 4 min **Score:** 1 pt **CLOs:** CLO 4

```
frontier = [(0, counter, start)]
while frontier:
    cost, _, current = heapq.heappop(frontier)
    if current in explored: continue
    explored.add(current)
    if current == goal: return path, cost
    for neighbor in env.get_neighbors(current):
        new_cost = cost + env.get_cost(current, neighbor)
        heapq.heappush(frontier, (new_cost, counter, neighbor))
```

Which are TRUE?

A) This implements Uniform Cost Search (UCS)

B) This implements A* Search with a heuristic

C) Goal test happens when node is **expanded** (popped)

D) With equal edge costs, this behaves like BFS

E) `counter` breaks ties in the priority queue

Part II. Probability & Bayesian Networks

Total Score: 11 points **Total Time:** 33 minutes

Question II.1: BN Construction & d-Separation (4 pts, 12 min)

Scenario: A company monitors network security with variables:

- `NetworkTraffic (N)`: High/Normal volume
- `IntrusionAttempt (I)`: Actual intrusion occurring
- `SystemLoad (S)`: Server under heavy load
- `FirewallAlert (F)`: Firewall generates alert
- `LogAnomaly (L)`: System logs show anomalies

Dependencies: I and S both cause high N. F depends on N and I. L depends on I and S.
Tasks:

(a) [2 pts] Draw the Bayesian Network structure.

(b) [1 pt] Are $I \perp S \mid F$? Use d-separation.

(c) [1 pt] Identify a collider and describe the "explaining away" effect.

Question II.2: Bayesian Inference (4 pts, 12 min)

Smart Home: $H \to M \to L$ (Home → Motion → Lights)

$P(H=T)$	0.6

$P(M\|H)$	$H=T$	$H=F$
$M=T$	0.9	0.1

$P(L\|M)$	$M=T$	$M=F$
$L=T$	0.95	0.05

Tasks:

(a) [2 pts] Compute $P(H=T \mid L=T)$ using enumeration.

(b) [2 pts] Apply Variable Elimination. Show factor tables.

Question II.3: Decision Under Uncertainty (3 pts, 9 min)

Drone Delivery: Weather: Clear (0.6), Cloudy (0.3), Storm (0.1).

Action	**Clear**	**Cloudy**	**Storm**
Fly Direct	+100	+60	−50
Take Detour	+70	+70	+30
Wait	+50	+50	+80

Tasks:

(a) [2 pts] Compute EU for each action: $EU(a) = \sum_s P(s) \cdot U(a, s)$

(b) [1 pt] Which action is MEU-optimal? Justify.

Part III. MDPs & Value Iteration

Total Score: 5 points **Total Time:** 15 minutes

Question III.1: MDP & Value Iteration (5 pts, 15 min)

Warehouse Robot: 2×3 grid. Goal G has reward +10. Living reward: −1/step. $\gamma = 0.9$.

S1	S2	S3
S4	S5	**G** (+10)

Transitions: 80% intended direction, 10% slip left, 10% slip right. Walls keep robot in place.
Tasks:

(a) [1 pt] Write MDP components (S, A, P, R, γ).

(b) [2 pts] Perform 2 iterations of Value Iteration for S3 and S5. Start with $V_0 = 0$. Use: $V_{k+1}(s) = \max_a[R(s,a) + \gamma \sum_{s'} P(s'|s,a)V_k(s')]$. Note: $V(G) = 10$.

(c) [2 pts] Extract optimal policy $\pi^*(s)$ for S3 and S5. Justify.

Part IV. Planning & Game Theory

Total Score: 6 points **Total Time:** 18 minutes

Question IV.1: Classical Planning — STRIPS (3 pts, 9 min)

E-Commerce Fulfillment: Predicates: `At(pkg,loc)`, `In(pkg,truck)`, `TruckAt(truck,dock)`, `Sorted(pkg)`.
Initial: At(P1,Intake), At(P2,Intake), TruckAt(T1,DockA), ¬Sorted(P1), ¬Sorted(P2)
Goal: In(P1,T1), In(P2,T1)
Tasks:

(a) [1 pt] Define STRIPS action `Load(pkg,truck,dock)`: Preconditions, Add List, Delete List.

(b) [2 pts] Show forward search plan to achieve goal. For each step: action applied, key state changes.

Question IV.2: Game Theory — Nash Equilibrium (3 pts, 9 min)

Tech Startup Competition: Two startups choose pricing: Premium, Budget, or Freemium.

A \ B	**Premium**	**Budget**	**Freemium**
Premium	(8, 8)	(3, 10)	(4, 9)
Budget	(10, 3)	(5, 5)	(6, 4)
Freemium	(9, 4)	(4, 6)	(7, 7)

Tasks:

(a) [1 pt] Identify any strictly dominated strategies. Apply IESDS if applicable.

(b) [2 pts] Find all pure-strategy Nash Equilibria. Mark best responses for each player.

Part V. Project Code Analysis

Total Score: 5 points **Total Time:** 12 minutes

Question V.1: Search Algorithm Analysis (3 pts, 6 min)

```
def astar(env, start, goal, heuristic='manhattan'):
    h = lambda pos: env.manhattan_distance(pos, goal)
    g_score = {start: 0}
    f_score = {start: h(start)}
    frontier = [(f_score[start], counter, start)]

    while frontier:
        current_f, _, current = heapq.heappop(frontier)
        if current in explored: continue
        explored.add(current)
        if current == goal: return path, g_score[current]
        for neighbor in env.get_neighbors(current):
            tentative_g = g_score[current] + env.get_cost(current, neighbor)
            if neighbor not in g_score or tentative_g < g_score[neighbor]:
                g_score[neighbor] = tentative_g
                f_score[neighbor] = tentative_g + h(neighbor)
                heapq.heappush(frontier, (f_score[neighbor], counter, neighbor))
```

Tasks:

(a) [1 pt] What do `g_score`, `f_score`, and `h()` represent? Explain $f(n) = g(n) + h(n)$.

(b) [1 pt] Why check `if current in explored: continue`? What if removed?

(c) [1 pt] Is Manhattan distance admissible for 4-connected grid? Why/why not?

Question V.2: Bayesian Belief Updates (2 pts, 6 min)

```
def update_single_cell_belief(prior, sensor_reading, sensor_accuracy=0.9):
    if sensor_reading:  # Obstacle detected
        likelihood_h = sensor_accuracy       # P(detect | obstacle)
        likelihood_not_h = 1 - sensor_accuracy  # P(detect | free)
    else:  # Clear
        likelihood_h = 1 - sensor_accuracy   # P(clear | obstacle)
        likelihood_not_h = sensor_accuracy   # P(clear | free)
    posterior, _ = bayes_update_full(prior, likelihood_h, likelihood_not_h)
    return posterior
```

Tasks:

(a) [1 pt] If prior=0.3, sensor_reading=True, accuracy=0.9, compute posterior. Show Bayes' rule.

(b) [1 pt] What happens if accuracy=0.5? Explain intuitively.